Young's Double Slit Experiment

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Abstract

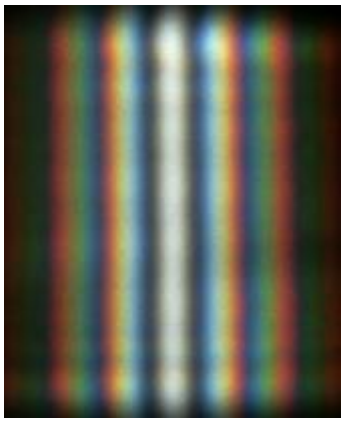
Theories on the composition of light have been numerous throughout the history of physics, with a significant milestone being the results from Young's Double Slit experiment, conducted in 1801. In this report, we present our own recreation of the experiment, and use the results to calculate properties of the light source including its wavelength. The wavelength we report is 702 ± 51 nm, a range within which the actual wavelength of 685 nm for red light lies.

Introduction

The famous Double Slit Experiment first performed by British polymath Thomas Young in 1801 led to fundamental changes in the understanding of the nature of light. Prior to the 18th century, the prevailing idea was the corpuscular theory of light (Descartes, 1637 and Newton, 1672) which was challenged by the wave-front theory (Huygens, 1678). Since the corpuscular theory could not explain the concepts of refraction, diffraction, interference and polarisation, Young's experiment provided a much-needed resolution to the debate. The conclusion that light could behave as a wave was a crucial insight that would later be built upon in 20th century with the founding of quantum mechanics (Plank, 1900).

In this modern recreation of the experiment, we used a laser as a source of directed monochromatic light precisely fired at diffraction gratings to produce an interference pattern on a screen a measured distance away. It is without doubt impressive that Young produced valid results with only candles and microscope slides in a darkened room.

Figure 1: A photo[1] depicting what Young would have seen in his original experiment, showing the chromatic dispersion due to the use of white light



THEORY

Light is an instance of electromagnetic radiation: an electric field and a magnetic field oscillating at right angles to each other to form a transverse wave that propagates through space without need for a physical medium. When a wave of wavelength λ passes through a slit of size *d*, the wave is diffracted, causing the wavefronts to spread out in a semi-circular pattern if *d* and λ are on a similar order of magnitude. Once diffraction has occurred, the intensity of the wave is distributed across two dimensions, in the plane perpendicular to both the original beam of light and the slit. A model for the extent to which waves are diffracted is given by Fraunhofer diffraction equation for a single slit,

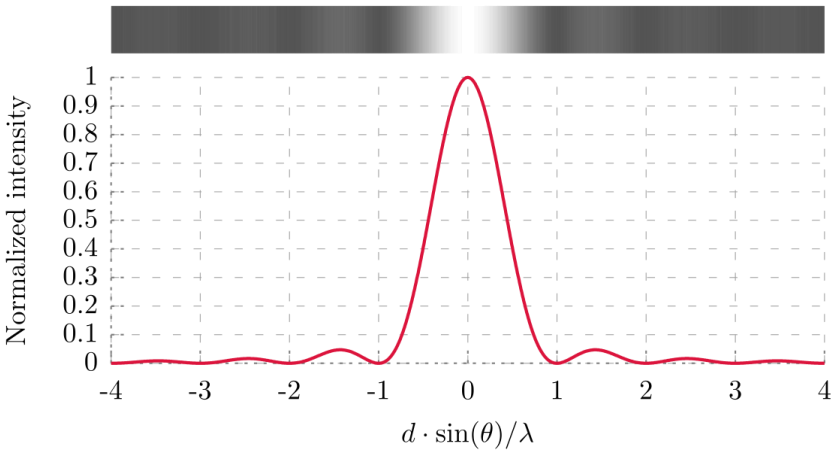
$$I(\theta) = I_0 \operatorname{sinc}^2\left(\frac{d\pi \sin(\theta)}{\lambda}\right) \qquad \text{(Equation 1)}$$

where *sinc*(*x*) is the 'unnormalised cardinal sine function', defined as

$$\operatorname{sinc}(x) = \begin{cases} 1, & \text{for } x = 0; \\ \sin(x)/x, & \text{otherwise} \end{cases}$$

which returns the intensity of the wave after diffraction at an angle θ, given the initial intensity I_0 and the slit size *d*. A plot of Equation 1 is shown in Figure 2.

Figure 2: A graph[3] of $\operatorname{sinc}^2(\pi x)$*, representing the intensity of a diffracted wave across a range with x being proportional to perpendicular distance, and an illustration of the light*



Equation 1 is independent of the radial distance, and hence the distance at between the first slit and the double slit is not important: we only require that the distances to each slit are equal, which we have assumed is the case.

When the diffracted waves pass a double slit of separation *a*, the wave is diffracted again, this time from two points. This creates an interference pattern as the waves from each point superpose. Since all parts of the wave travel at the same speed, the resulting wave is dependent only on the path difference between the two waves.

When the path difference is zero, this means both waves have travelled the same distance, and will be perfectly in phase since the waves are identical, creating a wave with double the original amplitude. This occurs along the line parallel to the original beam and is responsible for the brightest fringe in the centre of the detector screen. When the path difference is an even multiple of π (2π, 4π, etc), the same result occurs, along a different line with lower intensity. In contrast, when the path difference is an odd multiple of π (π, 3π, etc), the waves have travelled a distance such that at the point of superposition, all peaks line up with troughs and vice versa, leading to destructive interference – no wave is produced, leaving a dark fringe on the detector screen.

The distance between these fringes can be used to calculate the wavelength of the light, using the equation

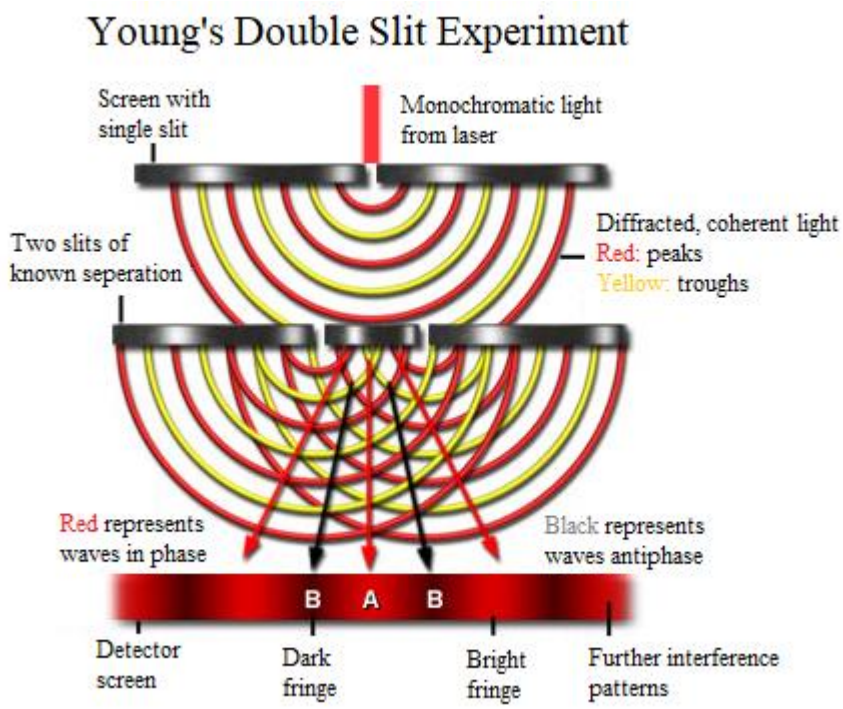
$$\lambda = \frac{ax}{d} \qquad \text{(Equation 2)}$$

where λ is the wavelength of the light, *a* is the separation between the double slits, *x* is the distance between the bright fringes and *d* is the distance between the double slits and the detector screen. All quantities are measured in metres. A derivation for Equation 2 is provided in the Appendix.

Practical Setup and Method

A diagram of our setup is shown in Figure 2.

Figure 2: Modern setup of Young's Double Slit Experiment[2]



The distance of the screen *d* was varied from 2.5 m to 5 m in 0.5 m intervals (plus an additional measurement at *d* = 6.17 m, the longest feasible distance) and the fringe separation *x* was measured each time. Throughout this practical, the same laser with the same wavelength of light λ was used and the same slits with separation *a*. The given value of λ was 685 nm while *a* was 1 mm. Once results were collected, a graph of *ax* against *d* was plotted. The gradient of the regression line represents our value for the wavelength λ.

Errors and Uncertainties

Sources of error and uncertainty in this practical may have come from:

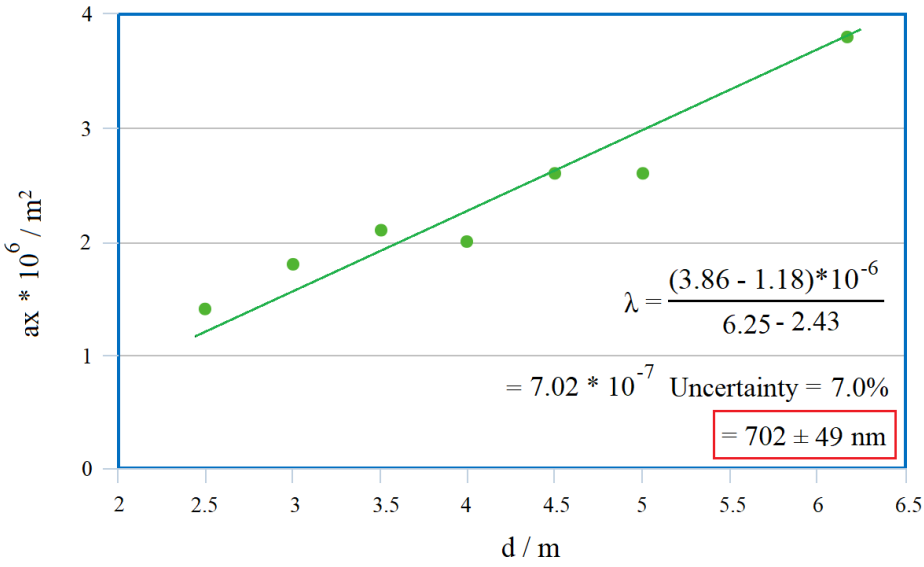
1. Measuring the distance *d*: Several metre rulers were used; despite a resolution of 1 mm, we only allow a minimum of ± 5 cm due to carelessness in the forms of accidentally moving the rulers/screen during the experiment and not making the rulers parallel to the laser. This produces errors in the range of 1-2%.
2. The slit separation *a*: We used a travelling microscope to determine that *a* = 1.00 ± 0.01 mm, a 1% error bound. A larger value of *a* would decrease the percentage uncertainty but make it more difficult to see the fringes and therefore increase uncertainty in *x*.
3. The fringe separation *x*: Instead of measuring only the distance between two fringes which would have given an extremely large uncertainty, a total distance across the 10 fringes was taken instead and divided by the number of fringes. The length has uncertainty ± 1 mm since it was measured with a ruler. We also gave a ± 1 uncertainty in the number of fringes as they were very small, numerous and difficult to count accurately. This produces errors in the range 3-7%.
4. Another potential source of error lies within the diffraction grating itself. If the single and the double slits were not exactly parallel, the diffraction would not accurately depict the correct interference pattern. However, we were unable to determine whether this is the case, and if so, it would be difficult to quantify the resulting uncertainty and therefore have chosen to ignore it (a 0% error).

From the above analysis, the main source of error in this experiment is the fringe separation. This error could be reduced by measuring the distance across more than 10 fringes, but this causes further problems as the more distant fringes are fainter and can be blurred, creating further uncertainty. In addition, the fringes are so small that it is difficult to accurately count the number of fringes when there are more than 10. Therefore, we believe that the total accumulated error (between 4-10%) has been made as small as practically possible with the given setup.

Calculations

Figure 3 shows a graph of *ax* against *d*. Error bars are omitted and were analysed in a seperate Excel spreadsheet. The worst-fit uncertainty in *ax* was 5.7% and in *d* was 1.3% giving a total uncertainty of 7%. The gradient of the best fit line was found to be $7.02 * 10^{-7}$ giving a base wavelength of 702 nm. Factoring in the error, we get that our result lies within the range 702 ± 49 nm.

Figure 3: Graph of results with line of best fit and summarised calculations



Given that the speed of light is $3.00 * 10^8$ ms^{-1}, we can use the formula $v = f\lambda$ to further calculate that the frequency of the red light was $4.27 * 10^{14}$ Hz with an uncertainty of ± $3.00 * 10^{13}$ Hz, which gives f = 427 ± 30 THz. According to an online source[4], the true frequency is 440 THz, so again this lies within our range. For this calculation, the speed of light in a vacuum *c* was used but correction for air is unnecessary as the refractive index of air[5] is 1.0003 so would not affect the result by any significant margin.

Conclusions

We obtained a final result of λ = 702 ± 49 nm = $(7.02 \pm 0.49) * 10^{-7}$ m, while the actual value was given as 680 nm, the reference wavelength for red light.

We claim the above results are valid because all errors have been quantified and accounted for, and the true value lies within these permitted error bounds.

Significance

Young's Double Slit Experiment proved the wave-like nature of light, a result that is fundamental to contemporary physics and has found application in the field of optics. A mechanism for diffraction proposed under quantum theory is only valid when light is modelled as a wave, demonstrating its salience in both pure and applied physics today.

The ability to measure wavelength, coupled with results of frequency-dependent experiments such as the photoelectric effect, allow determination of a value for the speed of light itself. Since the speed of light is a constant defined in terms of natural units, this in turn provides validation for a wide range of concepts involving light, such as Maxwell's Equations, the basis for electromagnetism, and Einstein's famous $E = mc^2$, which describes energetics at the nuclear level.

APPENDIX

A proof for Equation 2 is provided here.

Consider two light waves initially diffracted from two slits A and B such that the slit separation distance AB = *a* and the midpoint of AB is O. Let point Q be the brightest fringe on the screen such that OQ = *d* and P be the second brightest fringe. This is summarised in *Figure 4* below.

Since AQ = BQ and P and Q are both bright fringes, we know the path difference | BP – AP | = λ due to the constructive interference effects. Furthermore, let angle POQ = θ.

Next, if we assume a << d, the two light beams shown in red in Figure 4 become almost parallel. Therefore, there now exists a point on BP, call it C, such that AP = CP and hence by similar triangles, BAC = POQ = θ and BCA = 90°. This is summarised in Figure 5.

Figure 4

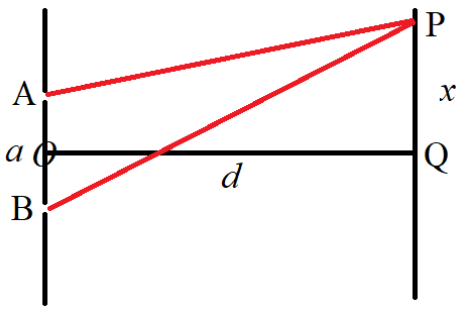
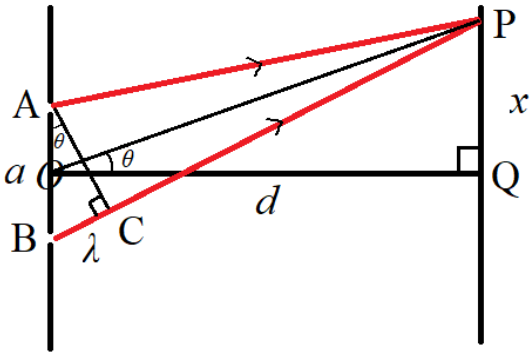


Figure 5



Using trigonometry in right-triangle ABC, we have

$$\sin(\theta) = \frac{\lambda}{a},$$ (Equation 3)

while in OPQ, we have

$$\tan(\theta) = \frac{x}{d}$$ (Equation 4)

Since x << d and a << d, we see that θ ≈ 0, so the small angle approximations are valid. Now, under these conditions we have sin(θ) = θ and tan(θ) = θ, which implies sin(θ) = tan(θ) allowing us to equate Equation 3 and Equation 4. We then get

$$\frac{\lambda}{a} = \frac{x}{d}$$

which, on isolation of λ, yields

$$\lambda = \frac{ax}{d}$$

as required by Equation 2.

References

[1] Photo by Aleksandr Berdnikov, Wikipedia

https://commons.wikimedia.org/w/index.php?curid=53009002

[2] Own work, image adapted from

https://micro.magnet.fsu.edu/primer/java/interference/doubleslit/

[3] Image by Wikipedia user DL6ER

https://commons.wikimedia.org/w/index.php?curid=34610901

[4] Frequency of Red and Violet Light, Quora

https://www.quora.com/Can-anyone-calculate-the-frequency-of-red-and-violet-light

[5] Refractive Index of Air, Britannica

https://www.britannica.com/science/refractive-index